

Strong–Weak Adaptive Cutoff Transfer for Intrinsic Riesz Factors

Sandwiched Functional Calculus, Midpoint–Ulam Contour Inheritance, and a Fixed-Window Route No-Go

Liang Wang

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Abstract

The factor-aware intrinsic-identification theorem for a small-noise quadratic Markov family left one analytic premise: a row-normalized adaptive Gaussian cutoff must remain small after the finite matrix recomputes its own Perron and parity Riesz factors. A direct Hilbert-space resolvent argument cannot close this premise. Endpoint spikes force the complete fixed-contour L^2 resolvent to grow at least as $\sqrt{\log(1/\sigma)}$.

We bypass that obstruction with a two-sided transfer identity. If a contour Γ avoids zero, then

$$\Pi(T; \Gamma) = \frac{1}{2\pi i} \int_{\Gamma} z^{-2} T(z - T)^{-1} T dz, \quad \mathcal{W}(T; \Gamma) = \frac{1}{2\pi i} \int_{\Gamma} z^{-1} T(z - T)^{-1} T dz.$$

The two outer transfer operators convert a strong-space contour resolvent into an $L^1 \rightarrow L^2$ Riesz bound. For reference and perturbed operators, the sandwiched defect is bounded by

$$D \leq \rho M R + S M \widetilde{M} \tau_1 R + S \widetilde{M} \tau_0,$$

where $M, \widetilde{M}$ are strong-space resolvent bounds, $R = \|T\|_{L^1 \rightarrow \mathcal{B}}$, $S = \|\widetilde{T}\|_{L^1 \rightarrow L^2}$, $\rho = \|T - \widetilde{T}\|_{L^1 \rightarrow L^2}$, and τ_0, τ_1 are the $L^1 \rightarrow \mathcal{B}$ and $\mathcal{B} \rightarrow \mathcal{B}$ defects.

We first bridge the exact Ulam operator and the full discretely normalized midpoint family. Conditioned folded-Gaussian derivatives give row L^1 error $O(h^2 \sigma^{-2})$ and piecewise-BV error $O(h \sigma^{-2})$. Hence the established strong-space peripheral contours pass to the midpoint family on every $h = o(\sigma^2)$ schedule. For a cutoff $L_{\kappa}(h) = \max\{L_0, \sqrt{2\kappa \log(1/h)}\}$, a distribution-free row-mass argument yields the sufficient threshold $\kappa \geq 7/4$. Using the Gaussian tail shape sharpens the Riesz defect to

$$\|\Delta \Pi\| + \|\Delta \mathcal{W}\| = O(h^{\kappa} \sigma^{-5/2}),$$

and lowers the sufficient all-schedule threshold to $\kappa \geq 5/4$. The archived choice $\kappa = 2$ therefore gives $o(\sigma^{3/2})$ on every $h = o(\sigma^2)$ schedule. Fixed L fails to vanish through this strong–BV route; this is not a theorem that the actual projectors diverge.

A four-level midpoint–Ulam audit and the five-scale intrinsic-factor data from the preceding paper support the mechanisms. A separate 256-bit Arb run audits the scalar ledgers. These computations are not production-scale interval eigensolvers. The result closes the analytic adaptive Riesz/cutoff premise, while the dyadically uniform Hardy–Stein trace budget remains open.

Keywords: Riesz projection; strong–weak stability; Ulam approximation; Gaussian cutoff; nonnormal perturbation; small noise.

MSC 2020: 47A55; 37M25; 47B65; 65R20; 37A30.

1 Introduction

The small-noise intrinsic determinant program has now reduced its sparse/full interface to a precise nonnormal question. RH-53 gave a deterministic all-column Hardy-tail certificate and the exact-real adaptive Gaussian matrix bridge [10]. RH-54 then propagated operator, coupling, projector, and weighted-Riesz defects through normalized Hardy triples and growing horizons [8]. The only missing input in that composition was an analytic modulus for the matrix’s *recomputed* intrinsic Riesz factors.

The naive solution is unavailable. For a common L^2 contour one may use two complete resolvents and the operator defect, as in the original weighted-Riesz perturbation theorem [12]. However, the endpoint square-root densities imply

$$\|\Pi_{\pm,\sigma}\|_{L^2 \rightarrow L^2} = \Theta\left(\sqrt{\log(1/\sigma)}\right),$$

so a uniform $O(1)$ fixed-contour L^2 resolvent is impossible [11]. This is a structural obstruction, not a defect of the numerical method.

The strong-weak perturbation framework used to identify the Perron and parity branches provides a different route. The conditioned folded-Gaussian transfer operator acts on a postcritical tower/spike space $\mathcal{B}$, with L^1 as weak norm and fixed isolating contours inherited from the deterministic branches [6, 9]. Gaussian smoothing then maps weak inputs to L^2 . The only missing algebra is to place that smoothing on both sides of the resolvent. The identities in the abstract do exactly this.

The paper has four analytic contributions.

1. It proves the sandwiched functional-calculus identities and a three-term strong-weak perturbation bound for both the unweighted projector and weighted Riesz term.
2. It supplies the midpoint-Ulam comparison omitted from the previous strong-weak factor theorem, including the conditioned normalizer and a piecewise-BV strong estimate.
3. It converts the exact twice-the-tail row identity into a generic mass-only adaptive theorem with sufficient threshold $\kappa = 7/4$.
4. It exploits the actual Gaussian boundary shape to remove the artificial cell-jump loss, yielding the sharper threshold $\kappa = 5/4$ and an $o(\sigma^{3/2})$ defect for the archived $\kappa = 2$ rule.

Both thresholds are sufficient for their respective proof ledgers. Neither is asserted to be necessary for actual Riesz stability. Likewise, the fixed-window negative statement concerns this strong-BV proof route only.

Three evidence levels are kept separate.

Analytic theorem Kernel derivative bounds, midpoint-Ulam transfer, strong-space contour inheritance, and the sandwiched Riesz modulus.

Binary64 diagnostic

Four midpoint-Ulam grids and five inherited dense intrinsic-factor scales.

Arb arithmetic audit

256-bit evaluation of the scalar tail and sandwich formulas, not an interval eigensolver for the production matrix.

2 Operator conventions and strong–weak contours

Let $I = [0, 1]$, $u = u_c$ be the first quadratic band-merging parameter, and $f(x) = 1 - ux^2$. Put

$$\phi_\sigma(t) = \frac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}, \quad g_\sigma(x, y) = \phi_\sigma(y - f(x)) + \phi_\sigma(-y - f(x)),$$

$$Z_\sigma(x) = \int_0^1 g_\sigma(x, y) dy, \quad k_\sigma(x, y) = \frac{g_\sigma(x, y)}{Z_\sigma(x)}.$$

The density transfer operator is

$$(\mathcal{P}_\sigma q)(y) = \int_I k_\sigma(x, y) q(x) dx. \quad (1)$$

Its transpose is the Markov operator on observables. We work on the density side because the tower/spike space $\mathcal{B}$ embeds compactly into L^1 .

Normalize the strong norm so that $\|q\|_1 \leq \|q\|_{\mathcal{B}}$. The established uniform Lasota–Yorke estimate has the form

$$\|\mathcal{P}_\sigma^m q\|_{\mathcal{B}} \leq A\vartheta^m \|q\|_{\mathcal{B}} + B\|q\|_1, \quad 0 < \vartheta < 1, \quad (2)$$

with constants independent of small σ . Keller–Liverani stability then supplies disjoint fixed-geometry contours Γ_+ and Γ_- around the branches converging to 1 and -1 [3]. In particular,

$$\sup_{z \in \Gamma_+ \cup \Gamma_-} \|(z - \mathcal{P}_\sigma)^{-1}\|_{\mathcal{B} \rightarrow \mathcal{B}} \leq M \quad (3)$$

for all sufficiently small σ .

The L^2 analogue of (3) is false uniformly. The proof below never assumes it.

3 Sandwiched Riesz functional calculus

Let T be bounded on a Banach space, and let $\Gamma \subset \rho(T)$ be a positively oriented rectifiable contour whose interior avoids the spectral point zero. Define

$$\Pi(T; \Gamma) = \frac{1}{2\pi i} \int_\Gamma (z - T)^{-1} dz, \quad \mathcal{W}(T; \Gamma) = T\Pi(T; \Gamma).$$

Lemma 3.1 (Two-sided transfer identities). *If 0 lies outside Γ , then*

$$\Pi(T; \Gamma) = \frac{1}{2\pi i} \int_\Gamma z^{-2} T(z - T)^{-1} T dz, \quad (4)$$

$$\mathcal{W}(T; \Gamma) = \frac{1}{2\pi i} \int_\Gamma z^{-1} T(z - T)^{-1} T dz. \quad (5)$$

Proof. Because T commutes with its resolvent,

$$T(z - T)^{-1} T = z^2(z - T)^{-1} - zI - T.$$

After multiplication by z^{-2} , the last two terms are $-z^{-1}I - z^{-2}T$. Their contour integrals vanish because zero lies outside Γ . This proves (4). Multiplying the same identity by z^{-1} leaves $z(z - T)^{-1} - I - z^{-1}T$; its integral is $T\Pi(T; \Gamma)$ because the last two terms again integrate to zero. $\square$

We now state the abstract perturbation estimate in the scale needed below. Assume $\mathcal{B} \hookrightarrow L^1$ and let $T, \tilde{T}$ act consistently on L^1 and $\mathcal{B}$. Put $R_T(z) = (z - T)^{-1}$ and similarly for $\tilde{T}$.

Theorem 3.2 (Strong–weak sandwich perturbation). *Suppose, uniformly on $z \in \Gamma$,*

$$\begin{aligned} \|R_T(z)\|_{\mathcal{B} \rightarrow \mathcal{B}} &\leq M, & \|R_{\tilde{T}}(z)\|_{\mathcal{B} \rightarrow \mathcal{B}} &\leq \widetilde{M}, \\ \|T\|_{L^1 \rightarrow \mathcal{B}} &\leq R, & \|\tilde{T}\|_{L^1 \rightarrow L^2} &\leq S, \end{aligned}$$

and define

$$\begin{aligned} \rho &= \|T - \tilde{T}\|_{L^1 \rightarrow L^2}, \\ \tau_0 &= \|T - \tilde{T}\|_{L^1 \rightarrow \mathcal{B}}, & \tau_1 &= \|T - \tilde{T}\|_{\mathcal{B} \rightarrow \mathcal{B}}. \end{aligned}$$

Then

$$\|TR_T(z)T - \tilde{T}R_{\tilde{T}}(z)\tilde{T}\|_{L^1 \rightarrow L^2} \leq D, \quad (6)$$

where

$$D = \rho MR + SM\widetilde{M}\tau_1 R + S\widetilde{M}\tau_0. \quad (7)$$

Consequently

$$\|\Pi(T; \Gamma) - \Pi(\tilde{T}; \Gamma)\|_{L^1 \rightarrow L^2} \leq \frac{\text{len}(\Gamma)}{2\pi} \max_{z \in \Gamma} |z|^{-2} D, \quad (8)$$

$$\|\mathcal{W}(T; \Gamma) - \mathcal{W}(\tilde{T}; \Gamma)\|_{L^1 \rightarrow L^2} \leq \frac{\text{len}(\Gamma)}{2\pi} \max_{z \in \Gamma} |z|^{-1} D. \quad (9)$$

Proof. Add and subtract the two intermediate products:

$$\begin{aligned} TR_T T - \tilde{T}R_{\tilde{T}}\tilde{T} &= (T - \tilde{T})R_T T \\ &\quad + \tilde{T}(R_T - R_{\tilde{T}})T + \tilde{T}R_{\tilde{T}}(T - \tilde{T}). \end{aligned}$$

Because $\|q\|_1 \leq \|q\|_{\mathcal{B}}$, the $L^1 \rightarrow L^2$ outer bounds also apply to strong-space inputs. The first and third terms therefore have bounds ρMR and $S\widetilde{M}\tau_0$. The resolvent identity

$$R_T - R_{\tilde{T}} = R_T(T - \tilde{T})R_{\tilde{T}}$$

may equivalently be written with the resolvents interchanged; using the placement consistent with the displayed product gives the middle bound $SM\widetilde{M}\tau_1 R$. Integrating [equations \(4\) and \(5\)](#) proves the result. $\square$

Remark 3.3. The theorem is stated as an $L^1 \rightarrow L^2$ estimate. Since the physical interval has unit measure, $L^2 \hookrightarrow L^1$ contractively, so it is also an $L^2 \rightarrow L^2$ estimate. On a finite equal-cell space, the standard isometric density lift identifies the latter with the Euclidean matrix norm. Thus [equations \(8\) and \(9\)](#) are exactly the intrinsic defects consumed by RH-54. No complete global L^2 resolvent appears.

4 From exact Ulam to the midpoint family

Let $I_i = [ih, (i+1)h)$, $x_i = (i+1/2)h$, and let E_h be equal-cell conditional expectation. The exact Ulam density operator is

$$U_{h,\sigma} = E_h \mathcal{P}_\sigma|_{V_h}.$$

Its row-stochastic observable transpose has entries

$$(U_{h,\sigma}^{\text{obs}})_{ij} = \frac{1}{h} \int_{I_i} \int_{I_j} k_\sigma(x, y) dy dx. \quad (10)$$

The full midpoint matrix used in the numerical branch is

$$(P_{h,\sigma}^{\text{mid}})_{ij} = \frac{g_\sigma(x_i, x_j)}{\sum_m g_\sigma(x_i, x_m)}. \quad (11)$$

These are not identical and must not be conflated.

Lemma 4.1 (Conditioned folded-Gaussian derivatives). *For all sufficiently small σ ,*

$$\sup_{x \in I} \int_I (|\partial_{xx} k_\sigma(x, y)| + |\partial_{yy} k_\sigma(x, y)|) dy \leq C\sigma^{-2}, \quad (12)$$

$$\sup_{x \in I} \int_I |\partial_y k_\sigma(x, y)| dy \leq C\sigma^{-1}. \quad (13)$$

For a target tail beginning at distance $L\sigma$, the first-variation bound (13), including its boundary jump, improves to $C\sigma^{-1}e^{-L^2/2}$.

Proof. For an unconditioned Gaussian,

$$\int |\phi'_\sigma| = O(\sigma^{-1}), \quad \int |\phi''_\sigma| = O(\sigma^{-2}).$$

The y derivatives follow by folding and restriction. For x derivatives, the chain rule adds bounded factors f' and f'' . It remains to control Z_σ^{-1} . The physical image $f(I) \subset [1-u, 1]$ and the symmetric conditioning interval $[-1, 1]$ give $Z_\sigma \geq c_0 > 0$ uniformly for small σ (indeed $Z_\sigma^{-1} \leq 2 + o(1)$); moreover $Z'_\sigma = O(\sigma^{-1})$ and $Z''_\sigma = O(\sigma^{-2})$ after scaling the boundary Gaussian. Differentiating g/Z therefore preserves the same powers. Finally, the target variation of one Gaussian tail is its boundary height: integrating $|\phi'_\sigma|$ from $L\sigma$ to infinity gives $\phi_\sigma(L\sigma) = C\sigma^{-1}e^{-L^2/2}$. Folding changes only the constant. $\square$

Theorem 4.2 (Midpoint–Ulam strong transfer). *Uniformly for h/σ sufficiently small,*

$$\max_i \sum_j |(P_{h,\sigma}^{\text{mid}})_{ij} - (U_{h,\sigma}^{\text{obs}})_{ij}| \leq Ch^2\sigma^{-2}, \quad (14)$$

$$\left\| (P_{h,\sigma}^{\text{mid}})^* - (U_{h,\sigma}^{\text{obs}})^* \right\|_{\mathcal{B}_h \rightarrow \mathcal{B}_h} \leq Ch\sigma^{-2}. \quad (15)$$

Here $\mathcal{B}_h$ is the cellwise spike/BV lift with its inherited strong norm. Consequently, on every schedule $h = o(\sigma^2)$, the fixed strong-space contours of the exact Ulam family are also common contours for the full midpoint family, with uniformly bounded strong resolvents.

Proof. Apply composite midpoint quadrature first in the target variable and then in the source cell. The vector-valued midpoint remainder is bounded in row L^1 by Ch^2 times (12). Replacing the continuum normalizer by the discrete row sum is a quotient perturbation with the same order because both normalizers are bounded below. This proves (14).

Lift a row difference to a cellwise-constant signed density. Its L^1 part is (14); the conservative jump variation costs one factor h^{-1} . The spike atoms are finite and cell averaging is variation diminishing on their regular representatives [1, 4]. Hence (15) follows.

The right side tends to zero precisely when $h = o(\sigma^2)$. A Neumann series around the exact-Ulam strong resolvent then gives common contours and a uniform midpoint resolvent upper. This is the finite-space contour inheritance required by theorem 3.2. $\square$

5 Generic mass-only cutoff transfer

Let $\tilde{P}_{h,\sigma}^{(L)}$ be obtained from the full midpoint row by omitting entries outside the archived support window and renormalizing. In this and the next section, T and $\tilde{T}$ denote the density transposes of the full and cutoff row-stochastic matrices. Uniform row bounds therefore become L^1 density-operator bounds by Minkowski's inequality. If the omitted full-row probability is q_i , then the exact identity of RH-39 is

$$\|\Delta \text{row}_i\|_{\ell^1} = 2q_i. \quad (16)$$

Put $Q_h = \max_i q_i$.

Lemma 5.1 (Distribution-free cell lift). *Without using any smoothness of the omitted entries,*

$$\rho \leq 2Q_h h^{-1/2}, \quad (17)$$

$$\tau_0 + \tau_1 \leq C(Q_h + Q_h h^{-1}). \quad (18)$$

Proof. The cellwise density lift of a row has L^2 norm at most its L^1 mass divided by $\sqrt{h}$, proving (17). A signed cellwise-constant density with L^1 mass $2Q_h$ has total jump variation at most $4Q_h/h$. Adding the L^1 part gives (18). $\square$

Gaussian smoothing gives

$$R = O(\sigma^{-1}), \quad S = O(\sigma^{-1/2}). \quad (19)$$

The first bound is a strong BV/spike norm; the second is the standard $L^1 \rightarrow L^2$ kernel estimate. Since the strong contour resolvents are uniform, [theorem 3.2](#) and [lemma 5.1](#) yield the following theorem.

Theorem 5.2 (Mass-only adaptive Riesz modulus). *If $Q_h/h = o(1)$, the cutoff family inherits the two strong-space contours, and for either contour*

$$\|\Delta \Pi\| + \|\Delta \mathcal{W}\| = O\left(\frac{Q_h}{h\sigma^{3/2}}\right). \quad (20)$$

For

$$L_\kappa(h) = \max\{L_0, \sqrt{2\kappa \log(1/h)}\},$$

the RH-39 Mills bound gives

$$Q_h = O\left(\frac{h^\kappa}{\sqrt{\log(1/h)}}\right),$$

and hence

$$\|\Delta \Pi\| + \|\Delta \mathcal{W}\| = O\left(\frac{h^{\kappa-1}\sigma^{-3/2}}{\sqrt{\log(1/h)}}\right). \quad (21)$$

On every strict schedule $h = o(\sigma^2)$ this tends to zero if $\kappa \geq 7/4$. For the archived $\kappa = 2$ it is

$$O\left(\frac{h\sigma^{-3/2}}{\sqrt{\log(1/h)}}\right) = o(\sqrt{\sigma}). \quad (22)$$

Proof. In (7), the middle term dominates the generic ledger and has size $O(\sigma^{-1/2}(Q_h/h)\sigma^{-1})$. This proves (20). The tail formula follows from $e^{-L_\kappa(h)^2/2} = h^\kappa$ and Mills' ratio [7].

On a boundary power schedule $h \asymp \sigma^p$, $p > 2$, the polynomial exponent is $p(\kappa - 1) - 3/2$. Its limiting worst case as $p \downarrow 2$ is nonnegative exactly when $\kappa \geq 7/4$; the logarithmic factor gives decay at equality. Equation (22) is immediate. Moreover $Q_h/h = o(1)$ throughout this range, so the contour-inheritance premise in the theorem is automatic. $\square$

6 Gaussian shape refinement

The preceding theorem treats an omitted Gaussian row as an arbitrary signed cell vector. That creates the Q_h/h jump penalty. The actual cutoff difference is much smoother: away from two support boundaries it is a rescaled Gaussian tail plus a small multiple of the retained full row.

Lemma 6.1 (Shape-aware cutoff norms). *Let $L_h \geq L_0 > 0$ and assume h/σ is small. Then the full and row-renormalized cutoff midpoint operators satisfy*

$$\rho = O(e^{-L_h^2/2}\sigma^{-1/2}), \quad (23)$$

$$\tau_0 + \tau_1 = O(e^{-L_h^2/2}\sigma^{-1}). \quad (24)$$

The constants are independent of h , σ , and the moving integer support boundary.

Proof. For one row, split the difference into the omitted tail and the retained renormalization term. The tail L^2 density norm is bounded by the square root of a Gaussian-square tail, giving $Ce^{-L_h^2/2}\sigma^{-1/2}$ after weakening any additional Mills factor. Its target variation is the integral of $|\partial_y k_\sigma|$ beyond $L_h\sigma$, plus the two boundary jumps. By lemma 4.1, both are $O(e^{-L_h^2/2}\sigma^{-1})$. The retained term is $q_i/(1 - q_i)$ times the full row; RH-39 gives $q_i = O(e^{-L_h^2/2}/L_h)$, while the full row has L^2 scale $O(\sigma^{-1/2})$ and variation $O(\sigma^{-1})$.

The operator acts on densities, so its strong output norm is target variation. No source derivative of the moving support is required: one integrates these uniform row-BV bounds against the input density. The integer support boundary itself contributes only the two target jumps just estimated. Thus no $1/h$ factor is paid. $\square$

Theorem 6.2 (Gaussian shape-aware Riesz modulus). *If $e^{-L_h^2/2}\sigma^{-1} = o(1)$, the cutoff family inherits the two strong-space contours, and for either contour*

$$\boxed{\|\Delta\Pi\| + \|\Delta\mathcal{W}\| = O(e^{-L_h^2/2}\sigma^{-5/2})}. \quad (25)$$

For the adaptive $L_\kappa(h)$,

$$\|\Delta\Pi\| + \|\Delta\mathcal{W}\| = O(h^\kappa\sigma^{-5/2}). \quad (26)$$

Every strict $h = o(\sigma^2)$ schedule is therefore preserved if

$$\boxed{\kappa \geq \frac{5}{4}}. \quad (27)$$

For RH-39's $\kappa = 2$ rule,

$$\|\Delta\Pi\| + \|\Delta\mathcal{W}\| = O(h^2\sigma^{-5/2}) = o(\sigma^{3/2}). \quad (28)$$

Proof. Insert equations (23) and (24) and (19) into (7). The first term is $O(e^{-L_h^2/2}\sigma^{-3/2})$; the middle term is $O(e^{-L_h^2/2}\sigma^{-5/2})$; and the third is $O(e^{-L_h^2/2}\sigma^{-3/2})$. The middle term gives (25).

For $h \asymp \sigma^p$, $p > 2$, the exponent in (26) is $p\kappa - 5/2$. It is nonnegative uniformly as $p \downarrow 2$ when $\kappa \geq 5/4$; on a strict little- o schedule the threshold case tends to zero. For $\kappa = 2$, $h^2\sigma^{-5/2} = o(\sigma^{3/2})$ because $h/\sigma^2 \rightarrow 0$. At every $\kappa \geq 5/4$, the strong defect $h^\kappa\sigma^{-1}$ also tends to zero, so the contour-inheritance premise is automatic. $\square$

Remark 6.3 (Sufficient, not necessary). The values $7/4$ and $5/4$ are thresholds for two explicit proof routes. They are not lower bounds on the true cutoff needed by the folded-Gaussian Riesz projector. Cancellations or more anisotropic strong norms could improve them.

Proposition 6.4 (Fixed-window route no-go). *If $L_h \equiv L < \infty$, neither the mass-only nor the Gaussian-shape strong-weak ledger gives a vanishing full-kernel Riesz perturbation as $\sigma \rightarrow 0$. In the shape ledger the strong cutoff proxy grows as $e^{-L^2/2}\sigma^{-1}$ and the Riesz proxy as $e^{-L^2/2}\sigma^{-5/2}$.*

Proof. The Gaussian boundary height $e^{-L^2/2}$ is a fixed positive number. Substitute it in equations (24) and (25). $\square$

Remark 6.5. This is only a no-go for the present norm route. It does not assert that the actual fixed-window projectors fail to converge, and it does not contradict the extremely small fixed-eight-sigma defects on every archived finite grid.

7 Numerical mechanism audit

The first audit compares (10) and (11) at $\sigma = 0.08$. Each source-cell integral is evaluated by 12-point Gauss–Legendre quadrature. The calculation is binary64 and is not used to establish the theorem.

n	h	max row L^1	ratio to $h^2\sigma^{-2}$	max lifted BV	ratio to $h\sigma^{-2}$
32	$3.1250 \cdot 10^{-2}$	$5.99398 \cdot 10^{-2}$	0.39282	1.19324	0.24438
64	$1.5625 \cdot 10^{-2}$	$1.58147 \cdot 10^{-2}$	0.41457	0.33176	0.13589
128	$7.8125 \cdot 10^{-3}$	$4.01200 \cdot 10^{-3}$	0.42069	0.08382	0.06867
256	$3.9063 \cdot 10^{-3}$	$1.00868 \cdot 10^{-3}$	0.42307	0.02125	0.03481

Table 1: Binary64 midpoint–Ulam mechanism audit. The row defect follows the predicted second-order scale; the conservative strong theorem requires only the weaker $O(h\sigma^{-2})$ envelope.

We also reuse, by hash, RH-54’s dense full-versus-sparse intrinsic-factor audit. It has $n\sigma = 5.12$, dimensions 32, 64, 128, 256, 512, and declared cutoffs $L = 5, 6, 8$. For the intentionally stressed $L = 5$ family:

σ	n	Q_h upper	actual $\Delta\Pi$	actual $\Delta\mathcal{W}$	sum/shape envelope
0.16	32	$4.9655 \cdot 10^{-7}$	$1.8305 \cdot 10^{-8}$	$2.3658 \cdot 10^{-8}$	$1.3410 \cdot 10^{-3}$
0.08	64	$4.9655 \cdot 10^{-7}$	$2.7670 \cdot 10^{-8}$	$3.7163 \cdot 10^{-8}$	$3.6626 \cdot 10^{-4}$
0.04	128	$4.9655 \cdot 10^{-7}$	$2.4205 \cdot 10^{-8}$	$2.3020 \cdot 10^{-8}$	$4.7163 \cdot 10^{-5}$
0.02	256	$4.9655 \cdot 10^{-7}$	$3.2570 \cdot 10^{-8}$	$2.9979 \cdot 10^{-8}$	$1.1043 \cdot 10^{-5}$
0.01	512	$4.9655 \cdot 10^{-7}$	$2.8136 \cdot 10^{-8}$	$2.6251 \cdot 10^{-8}$	$1.6973 \cdot 10^{-6}$

Table 2: Inherited binary64 intrinsic-factor stress audit. The last column uses the unit-constant power envelope and is descriptive, not a rigorous constant comparison.

The maximum observed projector-plus-weighted defect is 6.49×10^{-8} . At $L = 6$ it is about 10^{-10} beyond the smallest dimension, and at $L = 8$ it reaches binary64 roundoff. This supports the mechanism but does not replace the strong-space theorem.

Strong-weak adaptive cutoff transfer for intrinsic Riesz factors

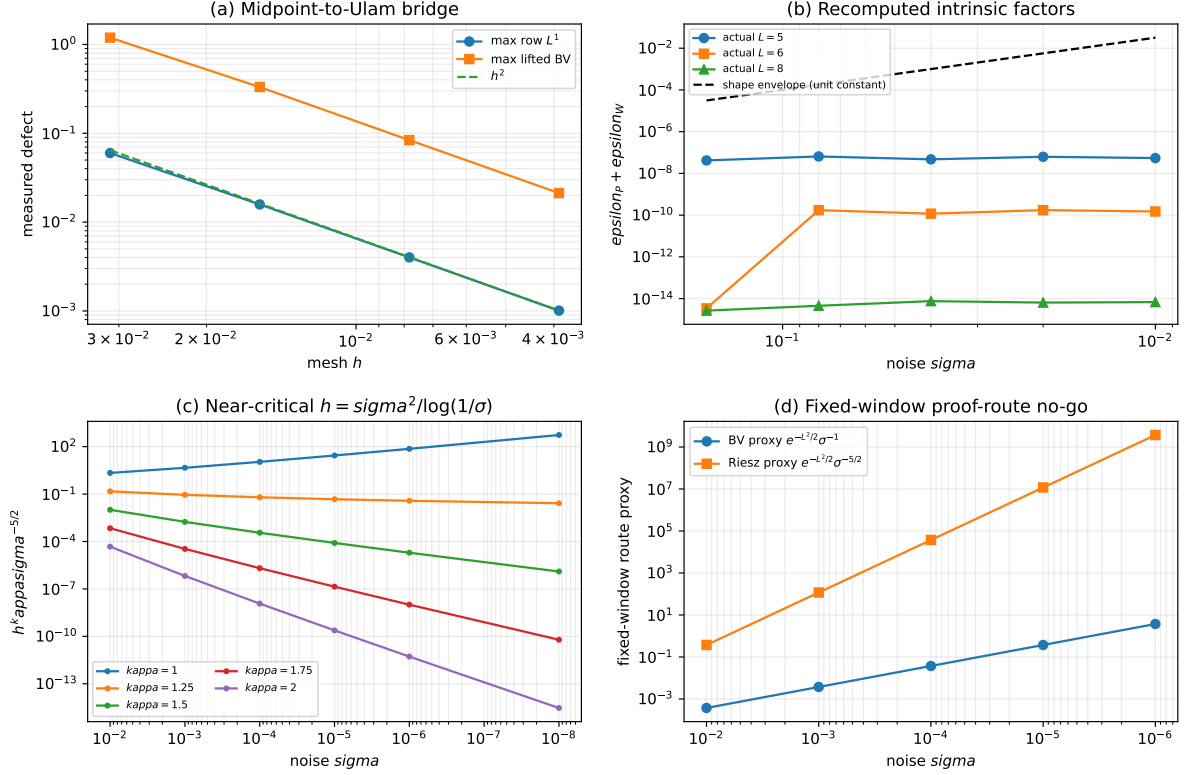


Figure 1: Strong-weak cutoff audit. (a) Midpoint-Ulam defects. (b) Recomputed intrinsic factors inherited from RH-54. (c) Shape-aware adaptive power envelopes on a near-critical mesh schedule. (d) Growth of the fixed-window strong-BV proof proxies.

7.1 Outward-rounded scalar ledger

A separate 256-bit Arb run evaluates the RH-39 omitted-mass formula at $(n, \sigma, L) = (512, 0.01, 5)$ and obtains

$$Q_h \leq 4.965522022961893 \times 10^{-7}.$$

It then composes interval inputs through all three terms of (7) and both contour weights. Finally, at the illustrative adaptive point $h = 10^{-12}$, $\sigma = 10^{-5}$, $\kappa = 2$, it certifies the unit-constant shape envelope divided by $\sqrt{\sigma}$ to be 10^{-9} . Arb guarantees outward arithmetic for these scalar formulas [2, 5]. The strong resolvent constants in that small audit are illustrative inputs; no production folded-Gaussian interval eigensolver is claimed.

8 Consequence for the program

RH-54's factor-aware theorem consumes separate Markov, unweighted-projector, weighted-Riesz, and normalized-coupling defects. The Markov and coupling parts were already controlled by RH-39, RH-53, and RH-54. The present theorem supplies the two missing intrinsic-factor defects on exactly the same adaptive family.

Corollary 8.1 (Adaptive factor-aware cutoff closure). *On every $h = o(\sigma^2)$ schedule, RH-39's adaptive choice $L(h) = \max\{5, 2\sqrt{\log(1/h)}\}$ gives vanishing full-versus-sparse unweighted and weighted peripheral Riesz defects. In fact their sum is $o(\sigma^{3/2})$. Therefore the analytic Riesz/cutoff premise in the RH-54 factor-aware intrinsic Hardy-triple transfer is satisfied.*

The resulting status is:

Midpoint–Ulam interface

Closed analytically on the adopted $h = o(\sigma^2)$ schedule.

Adaptive intrinsic Riesz modulus

Closed analytically for both projector and weighted term. No global uniform L^2 contour is used.

Stage A3 analytic factor transfer

Closed: the canonical full Gaussian family transfers to the adaptive sparse family through recomputed intrinsic factors.

Stage A3 production interval program

Not closed: the largest all-column traces and intrinsic factors have not been executed together in interval arithmetic.

Stage A1 Hardy–Stein budget

Open: no theorem yet gives a dyadically uniform/polylogarithmic trace budget for both directional Hardy energies.

Stage A4 intrinsic identification

Still conditional on Stage A1. This paper removes the Riesz/cutoff wall, not the Hardy wall.

The next focused question is therefore the uniform growing-horizon Hardy–Stein trace budget, without revisiting global L^2 conditioning.

8.1 No arithmetic or Hilbert–Pólya conclusion

Nothing here supplies an arithmetic trace formula, a von Mangoldt or prime-power identity, a zeta-zero spectral identity, a self-adjoint Hilbert–Pólya operator, or a $T \log T$ counting law. No Riemann-hypothesis conclusion is drawn. The independent TPC twin-prime branch is not an assumption in this argument.

9 Reproducibility and exact boundary

The archive contains the theorem algebra in `src/riesz_cutoff`, the midpoint–Ulam and inherited-factor pilot, the Arb scalar audit, machine- readable certificate, figure, tests, dependency hashes, and manuscript. Principal commands are:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_riesz_cutoff_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_arb_riesz_ledger.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_riesz_cutoff_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The sandwiched functional calculus, midpoint–Ulam scales, contour inheritance, adaptive exponent theorems, and fixed-window route no-go are analytic. The midpoint quadrature, eigensolver defects, and fitted visual slopes are binary64 diagnostics. The Arb execution validates

only the reported scalar arithmetic. These distinctions are part of the theorem statement, not implementation caveats.

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